Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K_4 Fe (CN)_6. After filtering, the precipitate is dried and weighed. The reactions which occur are ZnS + 2HCl \rightarrow ZnCl_2 + H_2 S 2ZnCl_2 + K_4 Fe (CN)_6 \rightarrow Zn_2 Fe (CN)_6 + 4 KCl If a 2 gram sample of rock yields 0.969 gram of Zn_2Fe(CN)_6, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn_2 Fe (CN)_6 = 342.6.

Answer Choices:

A. 10 % Zn

B. 22 % Zn

C. 20 % Zn

D. 19 % Zn

E. 15 % Zn

F. 18 % Zn

G. 16 % Zn

H. 14 % Zn

I. 12 % Zn

J. 25 % Zn

Reasoning + Answer:

<reasoning process here>

We are given:

- A 2 gram sample of rock yields 0.969 grams of ZnâĤĤFe(CN)âĤĨ.
- We are to find the percentage of Zn in the original rock sample.
- The reactions are:

1. ZnS + 2HCl âĨĴ ZnClâĤĤ + HâĤĤS

2. 2ZnClâĤĤ + KâĤĦFe(CN)âĤĨ âĨĴ ZnâĤĤFe(CN)âĤĨ + 4KCl

So, each mole of ZnâĤĤFe(CN)âĤĨ contains 2 moles of Zn.

We are given:

- Molecular weight of ZnâĤĤFe(CN)âĤĨ = 342.6 g/mol
- Atomic weight of Zn = 65.4 g/mol âĨĴ so 2 Zn atoms = 2 ÃĹ 65.4 = 130.8 g

Step 1: Find moles of ZnâĤĤFe(CN)âĤĨ produced:

moles = mass / molar mass = 0.969 g / 342.6 g/mol âīĪ 0.002829 mol

Step 2: Each mole of ZnâĤĤFe(CN)âĤĨ comes from 2 moles of Zn.

So, moles of Zn = 2 ÃĹ 0.002829 = 0.005658 mol

Step 3: Mass of Zn = moles ÃĹ atomic weight = 0.005658 ÃĹ 65.4 âīĪ ?

Compute:

0.005658 ÃĹ 65.4 =

First: 0.005 ÃĹ 65.4 = 0.327

0.000658 ÃĹ 65.4 âīĪ 0.0429

Total âīĪ 0.327 + 0.0429 = 0.3699 g

So, mass of Zn = 0.3699 g

Step 4: Percentage of Zn in the 2 g sample:

(0.3699 / 2) ÃĹ 100 = 18.495% âīĪ 18.5%

Closest option is 18%

Answer is F. 18% Zn

</reasoning process>

<answer>F</answer>